Dennis G. Pezan

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EDUCATION

United States Military Academy at West Point — West Point, NY
B.S., Operations Research (Finance Specialization), Minor in Artificial Intelligence — Expected May 2029

- CQPA: 3.571 | Dean's List, Fall 2025 and Spring 2026
- Research Project: Interdicting Taiwan's Crude Oil Supply (Advisor: LTC Matthew Webb)

RESEARCH EXPERIENCE

Contested Logistics / Network Interdiction Research – D/Math, USMA — West Point, NY
Undergraduate Researcher (Advisor: LTC Matthew Webb) — Jul 2026 – Present

- Extend LTC Webb's doctoral research on reinforcement learning for maximum-flow network interdiction to a 61-node, 83-edge open-source model of Taiwan's crude oil import network with nine costed adversary action types and 59 interdictable components
- Connected the network to a reinforcement-learning interdiction environment, verified it against an independent max-flow calculation, and fixed a targeting-filter issue that could hide the Strait of Hormuz
- Trained an agent for 39 hours whose base-case plan matches the exact optimum (Taiwan keeps 31.7% of demand), against random (about 90%) and greedy (70%) baselines; six sensitivity scenarios still to run

Reinforcement Learning for Contested Resupply – TDAC, U.S. Army — Fort Lee, VA
Research Intern, Modeling & Analysis Division (AIAD; Mentor: MAJ Erica Mikelson) — Jul 2026

- Redesigned the reward function for a proximal policy optimization (PPO) reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's stochastic sustainment simulation; trained and evaluated the agent in Python (PyTorch)
- Built two diagnostic tools that exposed four reward flaws, including an agent that outscored every baseline while starving the priority unit; the retrained agent more than tripled the share of requests filled and beat a hand-coded priority-aware baseline across 101 random seeds
- Presented findings to Branch Chief LTC Ryan Miller at Fort Lee and, by call, to TDAC's AI and reinforcement-learning researchers (LTC Joseph Pedersen) at the Naval Postgraduate School in Monterey, CA

Tree-Based Methods in Machine Learning – D/Math, USMA — West Point, NY
Independent Study in Machine Learning (MA289), Advisor: LTC Patrick Kuiper — Oct 2025 – May 2026

- Designed and completed an independent study of decision trees, bagging, and random forests, and presented the findings to officers
- Built an interactive random forest research site with a live classifier demonstration and hyperparameter explorer

PROJECTS

Data4Good Competition — Nationwide
Team Member — Oct 2026 – Present

- Team currently ranks first in a 13-state region; the regional winner presents at Johns Hopkins University

Tree-Based Methods Research Site – dennis-pezan.com/ma289.html — 2026

- Trained and evaluated decision trees, bagging, and random forests on the Titanic dataset in Python (scikit-learn) on an identical 80/20 split
- Computed pairwise tree correlation, overfitting gaps, and feature importance to show how ensemble decorrelation reduces variance; deployed via GitHub and Vercel

CFT Packing List – dennis-pezan.com/cft.html — 2026

- JavaScript web tool that helps cadets organize and track gear for Cadet Field Training; saves progress in the browser and deploys on Vercel from GitHub

Reward Engineering Case Study – dennis-pezan.com/rl-reward.html — 2026

- Public write-up diagnosing and correcting reward hacking in a simulated shipping network and raising requests met from 24% to 81%

Mounger Writing Center Website Redesign – dennis-pezan.com/mwc-redesign.html — Aug 2026 – Present

- Independent redesign concept for West Point's writing center site, built in Next.js, TypeScript, and Tailwind CSS with a custom WebGL ink shader and editable content files for staff

Professional Development

S2SS Technology and Innovation AIAD – SOSH, USMA California and Colorado
Summer AIAD on defense technology (separate from the academic-year Sea to Shining Sea Fellowship) Jun 2026

- Held in-depth discussions with leaders at Palantir, Stanford's Gordian Knot Center, OpenAI, Apple, and Meta Reality Labs on how defense technology is developed, procured, and fielded
- Observed how the 4th Infantry Division employs and gives feedback on these systems under Next Generation Command and Control; visited U.S. Northern Command, NORAD, and the Cheyenne Mountain Complex

Honors & Awards

Sea to Shining Sea Fellowship, United States Military Academy 2026–Present

- Academic-year fellowship with senior leaders in national security, government, finance, and technology; separate from the AIAD

Stokes Writing Fellowship, United States Military Academy 2026–Present

- Endowed rhetoric and literature fellowship that sponsors trips, conferences, and expos, with mentorship for publishing original writing

Norwegian Foot March Badge, Norwegian Armed Forces 2026
Dean's List, United States Military Academy Fall 2025, Spring 2026

Leadership & Service

C2 Claude Nationwide
Founder Feb 2026 – Present

- Founded and lead a startup of 18 cadets and graduate students working to build software together
- Established and administer an enterprise-level Claude environment, providing team members with shared AI capabilities to support software development and product creation
- Coordinate teams building consumer software, including a dating app, a multiplayer game, and design tools

Center for Enhanced Performance, USMA West Point, NY
Cadet-in-Charge, MA103 Data-Driven Modeling Tutoring Aug 2026 – Present

- Lead the full tutor staff for MA103, a core mathematics course with 1,010 cadets enrolled; direct tutoring coverage for every major graded event
- Built a study site (dennis-pezan.com/ma103-tutoring.html) with two full-length practice exams, a mirror of a graded gate, and a 59-drill reference card, each with an answer key

Mounger Writing Center, USMA West Point, NY
Writing Consultant Aug 2026 – Present

- Selected as a Stokes Writing Fellow to coach cadets on strengthening the rhetoric and argument of their writing

Center for Leader Development in STEM (CLD-STEM), USMA West Point, NY
Member Oct 2025 – Present

- Bring hands-on STEM projects to students, reaching 800+ students across the Chicagoland and Greater Los Angeles areas
- Selected to lead leadership and ethics workshops for 150+ Texas high school students at a West Point Leaders of Character Seminar

Cadet Ambassador Program, USMA West Point, NY
Cadet Ambassador Jan 2026 – Present

- Represent West Point in outreach to prospective cadets and help foster a welcoming environment for future applicants

Professional Experience

McDonald's Plainfield, IL
Department Manager Sep 2022 – Jul 2024

- Promoted at age 16 as the region's youngest department manager; sustained 90%+ customer satisfaction scores
- Scheduled 80+ employees weekly with 95% on-time accuracy; oversaw $20,000+ in weekly supply procurement

Skills

Technical: Python (PyTorch, scikit-learn), SQL, Java, JavaScript/TypeScript (Next.js) | **Methods:** reinforcement learning, network optimization, machine learning
Certifications: Microsoft Fabric Data Architecture (2026)
Languages: Spanish (fluent) | **Clearance:** Secret